
Which genes actually steer a T cell — and which just change the most?

An auditable method that separates effect size from functional direction and cross-donor repeatability in CD4+ T-cell Perturb-seq. The deliverable is not a target list — it is a tested scope map that knows when to call PASS, and when to stop.

Abel Costa · clinical research & hematology

A large effect can still be the wrong answer.

Some patients respond to a T-cell therapy, then relapse. To engineer better T cells we must tell apart the genes that control a cell's state from those that merely change when perturbed. A candidate controller must answer three questions:

REACH

How much did the cell move?

The size of the abnormal result — necessary, not sufficient.

PRECISION

Which state program moved?

Direction toward activation, Th1/Th2, memory, Treg or exhaustion.

REPEATABILITY

Did independent units agree?

Different guides and donors must reproduce the direction.

Magnitude alone is necessary — but not sufficient. Most screens stop at magnitude.

The useful question is: where did the T cell go?



M

Magnitude — how far the profile moved

C₁

Precision — movement along a frozen state axis

C₂

Repeatability — agreement across biological units

ISCI asks whether C adds information after M is known — the pre-specified $M \rightarrow M+C$ test.

PRIMARY RESULT

Direction and repeatability add signal beyond magnitude.

+0.357

**authoritative full-sample
M → M+C test**

95% CI [+0.117, +0.538]

+0.215

**leakage-free grouped
out-of-fold gain**

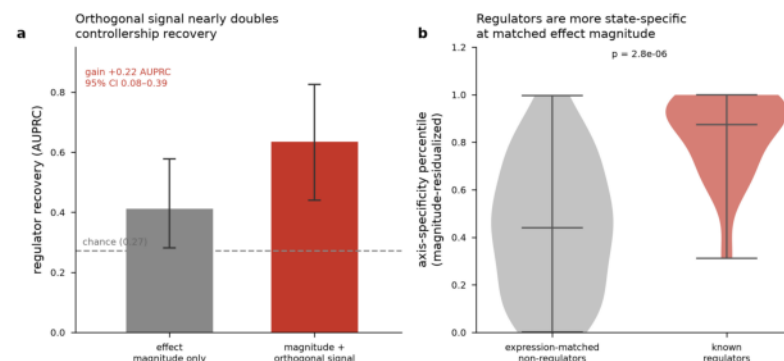
95% CI [+0.074, +0.560] · p = 0.010

0.415→0.722

**separate descriptive
ranking view**

AUPRC on the detectable set

These are distinct estimands — never averaged or swapped. With only 13 positives we re-ran everything out-of-fold and report the shrinkage.



TESTS DESIGNED TO BREAK THE RESULT

The signal is real — and biologically narrow.

We went looking for where it breaks.

External functional-regulator set

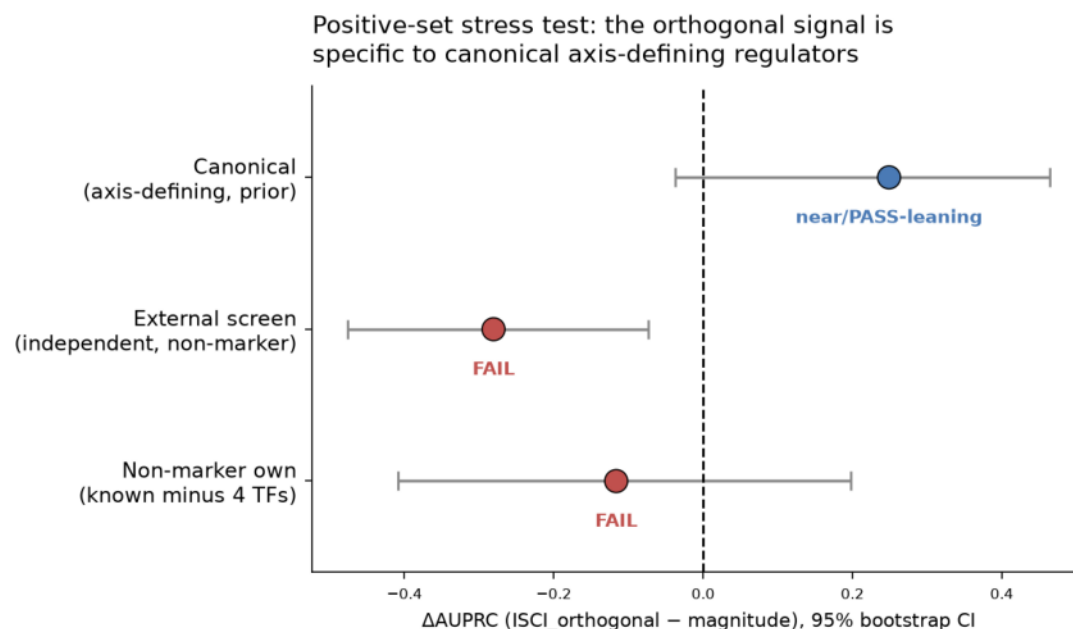
FAIL — ΔAUPRC -0.281 $[-0.476, -0.073]$

Interval sits entirely below zero.

Remove the four master TFs

GATA3, TBX21, STAT6, IRF1 $\rightarrow$ gain ≈ 0 (n=9)

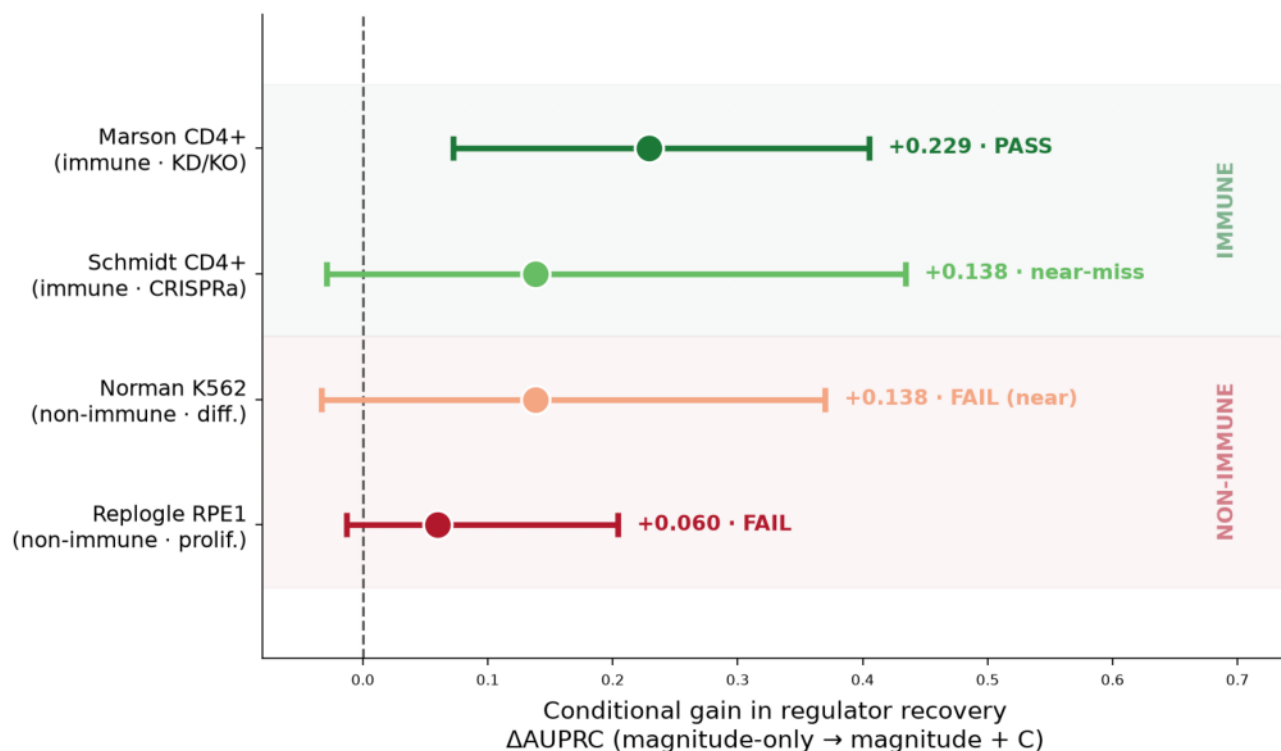
The claim narrows, honestly, to canonical axis-defining regulators in T cells.



External set = 20 non-marker T-cell functional regulators from published CRISPR screens (Shifrut/Schmidt/Belk), expression-matched negatives. FAIL is the honest bound: external functional regulators are magnitude-visible.

The property is immune-contextual, not universal.

CCI is immune-scoped — a graded boundary across 4 systems



PASS = bootstrap ΔAUPRC 95% CI excludes 0 AND conditional LR $p < 0.05$. Graded ordering matches TSC prediction P1: immune PASS > non-immune differentiation near-miss (Norman LR $p = 0.013$, carried by reproducibility R not axis-specificity S) > non-immune proliferation clean-FAIL (Replogle LR $p = 0.195$).

Marson CD4+

PASS

Schmidt / THP-1

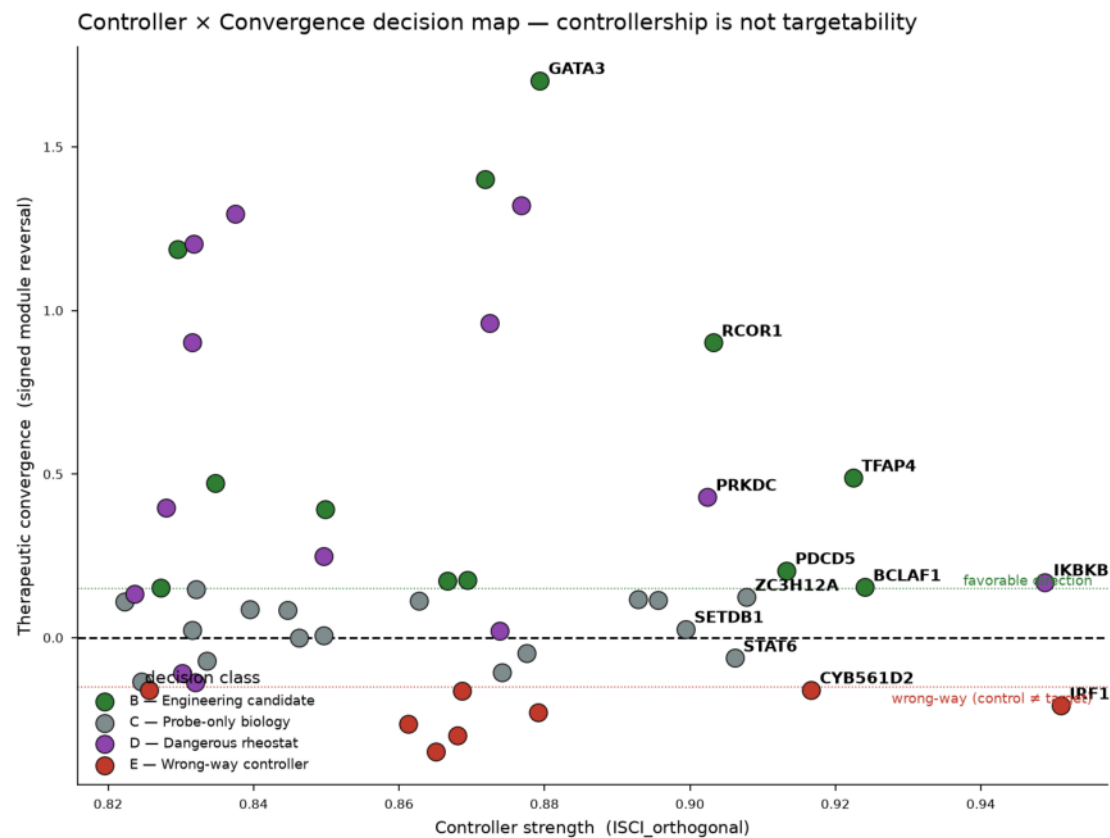
NEAR-MISS

K562 · RPE1

FAIL

A boundary is also a result. We do not use the word invariant.

Controller $\neq$ therapeutic target $\neq$ clinical biomarker.



Top-50 controllers. IRF1 is the #1 controller yet points the wrong way (negative convergence) — the concrete 'controller $\neq$ target' case. IKBKB/PRKDC are druggable but dangerous rheostats. Clinical relevance is capped at L1 (axis literature): the cross-study CAR-T response test was a well-powered NULL.

IRF1 ranks #1 as a controller

but points in an unfavorable intervention direction.

CAR-T response: NULL

study-out AUROC 0.533
CD8 baseline 0.585 beats it
87 patients · 9 studies

High controllership is not a promise of clinical benefit.

The product also knows when to stop.

PASS

Pre-specified direction holds and every required gate passes.

FAIL

The claim is testable and a required gate fails.

NULL

The test is valid but adds no supported signal.

NOT-EVALUABLE

A required input is absent; no biology is forced.

Every FAIL and NULL came from a correction loop: Claude challenged the headline, exposed a leakage path, forced the stricter test. The code owns the numbers; the skepticism is the science.

A prospective experiment that can falsify the next claim.

DONOR-RESOLVED DESIGN

A frozen ~54-guide CD4 panel across paired rest / stimulation contexts in 8–12 independent donors. The donor is the biological replicate — guides, wells and channels do not inflate n. Promotion requires concordant direction across the pre-declared panel.

FALSIFICATION

The extension fails if:

- residual precision gain below the frozen bound
- donor-bootstrap interval crosses zero
- gene-label / context-exchange $p \geq 0.05$

STOP BEFORE SYNTHESIS

Reference sequence, vector compatibility, on-target activity and off-target risk must pass first. A proposed panel is a design — not an experimentally validated result.

Executable, reusable, auditable.

1 · CONTRACTS

Frozen axes, claims, gates, DatasetSpec

2 · PYTHON ENGINE

CLI, adapters, block-bounded H5AD

3 · EVIDENCE

JSON/CSV, figures, hashes, tests, ledger

4 · EXPERIENCE

Notebook, offline HTML, video, deck

Run it on any compatible dataset: `uv run isci pipeline dataset.yaml`

21 / 21 automated release gates passed

MIT code · third-party data retain source terms · CITATION.cff · a tested scope map, not a target list